

ECESIS Investments

Summer 2026 Power Systems Modeling Internship Summary Report

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Assignment 1.

Introduction

This assignment introduces three data sources Dayzer, Panorama, and PJMISO market data. These sources provide constraint lists with different naming conventions and formats. The objective of this project is to match the constraints across the three data sources and output a csv file with the following format: (market_constraint, dayzer_constraint, pano_constraint).

Method

For this assignment a token based matching system was implemented. This strategy broke each constraint into its most basic units of information to find matches with the highest similarity. To prevent common words from dominating the matching score, the inverse document frequency (IDF) was used to build a dictionary relating words to their importance in matching.

Firstly, the data was loaded in and normalized by converting all separators to blank space, separating numbers from groups of letters, and removing duplicates. Voltage prefixes kv and l were removed to allow matching between different formatting. For market data, constraint and contingency columns were combined and for panorama data, monitored facility and contingency name were combined for added matching context.

Next, all constraints were given to TfidfVectorizer from SKlearn to find the IDF of each word as its importance for matching purposes. This technique gives higher weight to less frequently used words. When seldom used words match, there is a higher chance of the constraint corresponding to the same entity.

In an effort to reduce the computational cost of matching the entities, a token index was built for the Dayzer and Panorama datasets. This dictionary points to all index locations of the token key. This allows for efficiently finding token matches without searching the entire market dataset.

For each constraint in the market data, the match score was found by looping through the tokens in said constraint and querying the token index for Dayzer and Panorama to find the location of matches in the respective datasets. Each matching index was stored as a key in the scores dictionary with the word importance added to the corresponding value for each match. Once all tokens for the current market constraint were evaluated, the index with the highest match score was chosen as the best match to the current constraint.

Conclusion

The aforementioned method produces adequate results with spot checking indicating a high degree of accuracy in a run time of only 50 seconds. This method works well for matching constraints with the same spelling but may fall short on datasets with varying abbreviations or typos. There is a slight bias towards longer constraint names as they simply contain more tokens and therefore have a chance of having many coincidental matches that outweigh the score of a better match. A matching threshold was not defined, giving way to bad matches when no high quality matches are found. The final shortcoming of this strategy is that constraints are not removed from the matching pool after a match is found. This could result in multiple market constraints being matched to the same Dayzer or Panorama constraint.

Assignment 2.

Introduction

This assignment requests a comparison of two different forecasting techniques to predict bus level power demand. The first technique is a direct bus forecast feeding historical bus level load data into a machine learning model to predict next-day loads as well as next-month loads. The second method is projecting a zone level prediction to the bus level using historical bus share for each zone. LightGBM, a gradient boosting framework, was used to make predictions.

Method

This solution begins by combining previous years datasets and building predictive features such as day of week, day of year, month, a weekend flag, and lagged values including a 7 and 35 day as well as a 1 year lag. Due to the size of the data, duckdb was implemented to build the feature set without loading all values into memory. Two parquet files are then saved to the directory, one for the bus level features and one for the zone level features

An additional parquet file for bus share is created for projecting zone predictions to the bus level. This projection is lagged by 2 months to ensure only data available at the time of model training was used. For example, a model trained to forecast May demand will use March bus share since the monthly model will need to be trained in April and will not yet have access to the April bus share.

Once the features are created, every day of the forecast period is analyzed sequentially. The current day was treated as the prediction day. A next-day prediction model was retrained every 7 days on data gathered 2 to 37 days prior to the first day that it would be applied to ensure the model could have been trained the day before the first prediction. A next-month model was retrained before predictions were made for a new month. The same training window was applied to ensure no data leakage impacted the models predictions. The monthly model also omitted the 7 day lag to cease data leakage.

On each day of the prediction period, features from the day were given to the corresponding LightGBM model for next month and next day predictions. These forecasts were saved until the end of the month. At the end of each month, zone level predictions were converted into bus level loads if applicable. To conserve memory, the monthly predictions were temporarily stored on the disk to be combined at the end of the prediction period.

At this point statistics were gathered for each model for both the baseline which was evaluated as the load of the bus at the same time last year, and the model prediction. The statistics for each model are available in Appendix A.

Conclusion

This solution utilized the lightGBM gradient boosting framework to forecast next day energy demand as well as next month energy demand. Predictions were compared against the demand of the bus at the same hour one year prior to see if utilizing a machine learning model improved the accuracy of the forecast. This model works well at forecasting stationary bus loads but may lack the ability to accurately predict trending bus loads. This claim is supported by the increase in MAE in month 1 and 6 where weather is changing and bus loads may be trending as a result.

Both forecasting techniques resulted in lower error across all three metrics when compared to the baseline. The direct bus forecasting method exhibited a higher accuracy than the zone bus share forecasting method. This is likely due to the zone bus share method incurring error both in zone load predictions and in the projection of historical bus share. Since the bus share was lagged by two months, it was likely outdated by the time it was applied to the zone loads.

For the direct bus forecasting method, the next day model resulted in a lower error compared to the next month model. This result is possibly explained by the more frequent retraining of the next day model adjusting quicker to changing bus share.

The statistics for the zone bus share forecasting method showed comparable results among the next day and next month model. This result suggests that zone level predictions were similar since the same bus share was applied to both models. A possible explanation is that zone forecasts are more stable and therefore the more frequent retraining frequency of the next day model did not have meaningful improvement on the accuracy.

Future improvements for this pipeline include further research into features with predictive power and performance optimization to minimize runtime. A large portion of the runtime comes from retraining the next day model. The training window can be optimized for either high accuracy or decreased runtime.

Appendix A.

Direct Bus Forecast

Table 1: Next Day Prediction Model Statistics

month	MAE_predict	RMSE_predict	WMAPE_predict	MAE_baseline	RMSE_baseline	WMAPE_baseline
1	2.387764	6.862588	0.210059	3.831457	11.061389	0.317284
2	1.997500	5.641275	0.186147	3.554443	11.946355	0.312030
3	1.733506	6.757604	0.177911	2.911822	10.161469	0.280035
4	1.758620	4.302648	0.167017	3.188050	12.441107	0.282010
5	1.685194	4.957673	0.147046	3.557385	13.442819	0.291735
6	2.255285	6.072237	0.173790	3.531206	12.467493	0.256965
7	1.826088	4.848813	0.137657	3.754201	12.995378	0.267143
8	1.764010	4.844982	0.128654	3.425099	13.085479	0.236132
9	1.051570	3.384064	0.083521	3.110914	12.320311	0.232261
10	1.391907	3.799051	0.123172	3.104611	10.317633	0.256373
11	1.867218	4.978705	0.184183	3.214113	11.635947	0.296025
12	1.672469	6.760660	0.163482	3.197439	11.073547	0.293179

Table 2: Next Month Prediction Model Statistics

month	MAE_predict	RMSE_predict	WMAPE_predict	MAE_baseline	RMSE_baseline	WMAPE_baseline
1	3.072982	10.685086	0.270340	3.831457	11.061389	0.317284
2	2.091957	6.007758	0.194949	3.554443	11.946355	0.312030
3	2.091702	7.331263	0.214673	2.911822	10.161469	0.280035
4	2.104668	4.907865	0.199881	3.188050	12.441107	0.282010
5	1.846672	5.154164	0.161136	3.557385	13.442819	0.291735
6	3.042145	8.703941	0.234424	3.531206	12.467493	0.256965
7	2.056123	5.734839	0.154998	3.754201	12.995378	0.267143
8	1.897869	6.037065	0.138417	3.425099	13.085479	0.236132
9	1.274002	3.904094	0.101188	3.110914	12.320311	0.232261
10	1.455252	4.230499	0.128778	3.104611	10.317633	0.256373
11	2.642661	7.554669	0.260674	3.214113	11.635947	0.296025
12	1.952395	7.350890	0.190844	3.197439	11.073547	0.293179

Zone Bus Share Forecast

Table 3: Next Day Prediction Model Statistics

month	MAE_predict	RMSE_predict	WMAPE_predict	MAE_baseline	RMSE_baseline	WMAPE_baseline
1	3.202493	9.657744	0.279384	3.827095	11.055613	0.317327
2	2.911154	8.183439	0.271510	3.515136	11.825625	0.311670
3	1.514559	4.699234	0.155200	2.912370	10.163399	0.282347
4	2.333079	6.796741	0.219831	3.188529	12.447486	0.283182
5	2.351857	6.405204	0.203425	3.552337	13.446868	0.291842
6	3.042837	10.299496	0.232439	3.530580	12.479792	0.257579
7	3.231932	9.709095	0.242310	3.753289	12.998214	0.268070
8	2.454174	6.954441	0.177981	3.424345	13.092314	0.236521
9	1.567791	4.455053	0.124028	3.104410	12.314392	0.232654
10	1.350954	4.019025	0.119156	3.077920	10.248042	0.256455
11	2.478768	7.192641	0.242961	3.215951	11.642448	0.297930
12	2.478862	6.575262	0.240807	3.196615	11.079060	0.295190

Table 4: Next Month Prediction Model Statistics

month	MAE_predict	RMSE_predict	WMAPE_predict	MAE_baseline	RMSE_baseline	WMAPE_baseline
1	3.307801	10.473479	0.288571	3.827095	11.055613	0.317327
2	2.633725	8.054439	0.245636	3.515136	11.825625	0.311670
3	1.424294	4.404418	0.145950	2.912370	10.163398	0.282347
4	2.293119	6.740456	0.216066	3.188529	12.447486	0.283182
5	2.265065	6.148901	0.195918	3.552337	13.446868	0.291842
6	3.193350	9.102463	0.243936	3.530580	12.479792	0.257579
7	3.273732	9.332596	0.245444	3.753289	12.998214	0.268070
8	2.437118	6.921009	0.176744	3.424345	13.092314	0.236522
9	1.802427	4.776677	0.142590	3.104410	12.314392	0.232654
10	1.390526	4.066648	0.122646	3.077920	10.248042	0.256455
11	2.534224	7.650243	0.248396	3.215951	11.642448	0.297930
12	2.308224	6.678896	0.224231	3.196615	11.079060	0.295190